Chihuahua vs Muffin

Data-Centric AI Image Classification

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3LC × MNNIT Data-Centric AI Hackathon | March 2026

96.79%
Public Leaderboard Score

95.50%
Best Validation Accuracy

ResNet-18
Model (from scratch)

Competition Overview

The 3LC × MNNIT Data-Centric AI Hackathon challenged participants to build a binary image classifier distinguishing **chihuahuas** from **muffins**. The model architecture was **fixed as ResNet-18**, trained entirely from scratch (no pretrained weights). Improvement could only come from **data curation** using the 3LC platform.

Split	Labeled	Unlabeled	Total	Purpose
Train	100 (50+50)	3,579 undefined	3,679	Training pool
Validation	1,000 (balanced)	—	1,000	Monitor accuracy
Test (hidden)	—	—	1,184	Kaggle scoring

Results

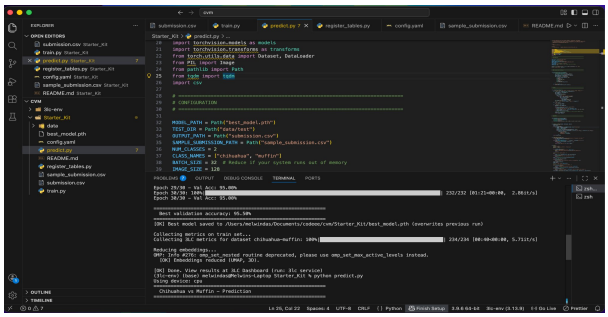

Fig 1: Training terminal — best val acc 95.50%

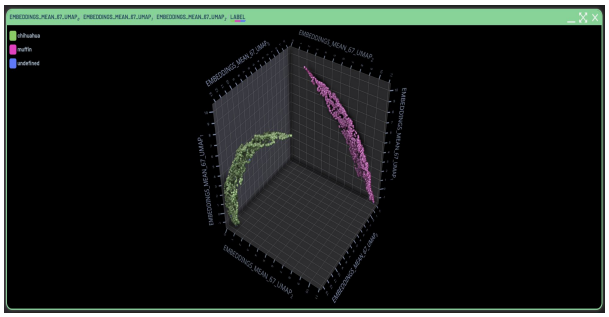

Fig 2: 3LC Dashboard — 3D UMAP embeddings (green=chihuahua, pink=muffin)

Our Approach

Data Labeling Strategy

We started with 100 labeled images and a pool of 3,579 unlabeled (undefined) images. Our labeling process used three complementary signals:

- **Confidence scores:** Sorted undefined samples by model prediction confidence. High-confidence predictions were reviewed first as they were the easiest to verify and label correctly.
- **Manual inspection:** Each candidate sample was visually checked before assigning a label. Some images were genuinely ambiguous — extreme close-ups, unusual angles, or low quality — so we chose not to include them in training.
- **Embedding clusters:** Used the 3LC Dashboard's 3D UMAP visualization to identify clusters of undefined samples. Samples sitting clearly inside a chihuahua or muffin cluster were labeled accordingly, making bulk labeling faster and more reliable.
- **Excluding confusing samples:** Images that were hard to classify even by eye were set to **weight = 0** in 3LC so they were excluded from training, preventing noisy labels from hurting model performance.

Training Modifications

The provided `train.py` was modified with the following changes:

- **Epochs increased to 30:** The default number of epochs was extended to 30 to allow the model more time to converge on the expanded labeled dataset.
- **Cosine Annealing inside StepLR:** Replaced the default scheduler with `CosineAnnealingLR(optimizer, T_max=30)` for smoother learning rate decay, helping the model settle into a better minimum over the full training run.